Gobinath Kulathunga Ashwinthan

Electrical and Electronic Engineering Undergraduate

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INTRODUCTION

Final-year Electrical and Electronic Engineering undergraduate, currently building skills in digital design, Verilog, and FPGA-based systems. Actively exploring VLSI, RTL design, and hardware verification through continuous hands-on projects and self-learning. Deeply interested in chip design and logic development, with growing experience in Verilog-based design and FPGA testing. Fluent in English, Tamil, and Sinhala. Seeking an internship to gain practical experience and contribute meaningfully in embedded systems, digital electronics, or semiconductor design roles.

TECHNICAL SKILLS

Programming: C, C++, MATLAB, Python, ARM Assembly, Verilog HDL
Libraries Used: NumPy, SciPy, Pandas, Matplotlib, TensorFlow, PyTorch, OpenCV, Scikit-learn
Microcontrollers: ESP32, Arduino (AVR)
Tools: Arduino IDE, Proteus, KiCad, Vivado Xilinx, PS-CAD

PROJECTS

8-bit ALU Design

Dec 2025

Verilog HDL, Digital Arithmetic Modules

Overview: Designed a fully structural 8-bit Gate-Level ALU supporting **signed** and **unsigned** arithmetic with a **ripple-carry two's complement adder–subtractor** and explicit overflow detection and multiplication using a **sequential shift-and-add architecture** and division via an **iterative restoring division algorithm** with shared subtract–compare logic. Integrated multi-cycle control (**start/busy/done handshake**) and a complete flag set: zero, negative, overflow, carry-out, and divide-by-zero.

Parameterized Synchronous FIFO in Verilog

Feb 2026

Verilog HDL, RTL Verification, Memory Architecture

Overview: Designed and verified a **parameterized single-clock synchronous FIFO** in Verilog RTL with configurable **depth** and **data width**. Implemented a **circular buffer** using phase-extended read and write pointers for reliable full and empty detection. Verified correct operation across **pointer wraparound** and **concurrent read/write conditions**. Developed a **task-based testbench** with directed scenarios and waveform analysis to validate data orderingand status flag behavior.

Sobel Edge Detection using Verilog

Feb 2026

Verilog HDL, FPGA, VGA Controller

Overview: Implemented a real-time streaming Sobel edge detection pipeline on Artix-7 (Basys-3) using Verilog RTL, processing **640×480 VGA at 25 MHz**. Designed **dual-BRAM line buffers** and a **3×3 window generator** to achieve one-pixel-per-clock convolution. Integrated RGB565 image ROM, fixed-point grayscale conversion, pipelined gradient computation, saturation, and dynamic threshold control. Demonstrated multiple **display modes** (RGB, grayscale, edge, overlay) with precise VGA timing and latency alignment.

Distance Detector Using Dual Loop Antennas

2024 – 2025

ESP32, ATMEGA328p, LoRa Modules, Dual Loop Antennas, AVR Assembly, C++

Overview: Developed an RF source tracking system using **null detection** with custom **dual loop antennas** based on the triangulation principle at **418MHz**. Enabled real-time direction finding by detecting signal nulls through antenna orientation.

Contributions: Designed and tested resonant dual loop antennas, and implemented AVR assembly code for sine lookup table generation and triangulation logic using unsigned fixed-point arithmetic.

Final-Year Project: Smart Crack Detection and Structural Monitoring System *2024 – Present*

ESP32, LoRa, Capacitive Sensors, IR Imaging, Python, CNN

Overview: Developing a **smart sensor network** for concrete structural health monitoring, exploring multiple sensing modalities with primary focus on **capacitive sensing**, and integrating **machine learning and image processing** for crack detection and propagation analysis.

Contribution: Designed and trained a grayscale-based **CNN crack detection model** using curated crack image datasets. Currently working on hardware integration of the image processing pipeline, with **Sobel edge detection validated on FPGA**, and progressing toward full **CNN deployment on FPGA** for on-chip crack inference.

Subsurface Emergency Transmitter using Near-Field Magnetic Induction *Oct 2025*

KiCad, PCB Design, Class-AB Amplifier, LC Tank, Boost Converter

Overview: Developed a 125kHz near-field magnetic induction transmitter for underground emergency signaling during landslides and disaster rescue missions. System design included a resonant LC tank, Class-AB amplifier, and energy-efficient power delivery stages.

Contributions: Designed the boost converter for magnetic field drive and completed full two sided PCB layout in KiCad, including component footprinting and hardware validation.

Image Compression using Huffman Coding *Jan 2026*

MATLAB, Custom Algorithms

Overview: Developed a grayscale image compression pipeline using Huffman coding, fully implemented from first principles in MATLAB without relying on built-in functions. Computed pixel intensity histograms, derived symbol probabilities, and generated optimized binary codebooks for entropy encoding. Encoded original and cropped images into compact bitstreams and stored them as text files for reconstruction and compression analysis.

EDUCATION

VLSI CAD: Logic (Part I) — Ongoing Online Course 2025 – Present

Coursera, University of Illinois Urbana-Champaign

B.Sc. Engineering (Hons.) in Electrical and Electronic Engineering (GPA - 3.279) 2022 – Present

University of Peradeniya

G.C.E. Advanced Level and Ordinary Level Examinations 2012 – 2021

Royal College, Colombo 07

Primary Education 2007 – 2011

Colombo Hindu College, Bambalapitiya

EXTRACURRICULAR ACTIVITIES

Karate - University of Peradeniya 2023

Sri Lanka University Games (SLUG 2023)

- Represented University of Peradeniya in Karate at the Sri Lanka University Games 2023, showcasing discipline, teamwork, and competitive spirit.

Diploma in Electronic Keyboard 2025 – Present

London College of Music (LCM), University of West London

- Pursuing a diploma in Electronic Keyboard through the London College of Music, refining musical expression and technical proficiency.

REFERENCES

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